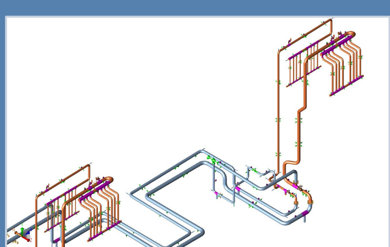
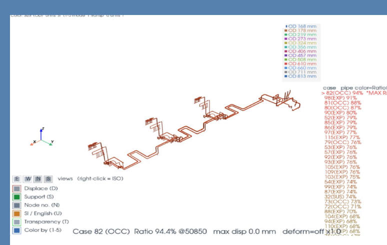
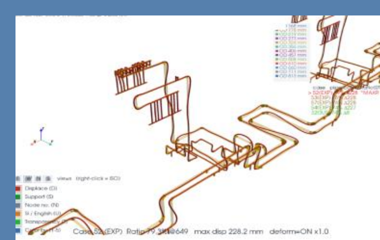
CAESAR II → E3D / SP3D Converter

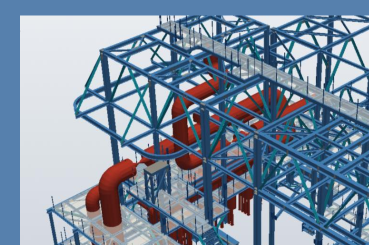
< Automation of 3D Piping Model from Stress Analysis >

CAESAR II Neutral (.CII) → E3D Macro · AP3D DXF · SP3D IFC









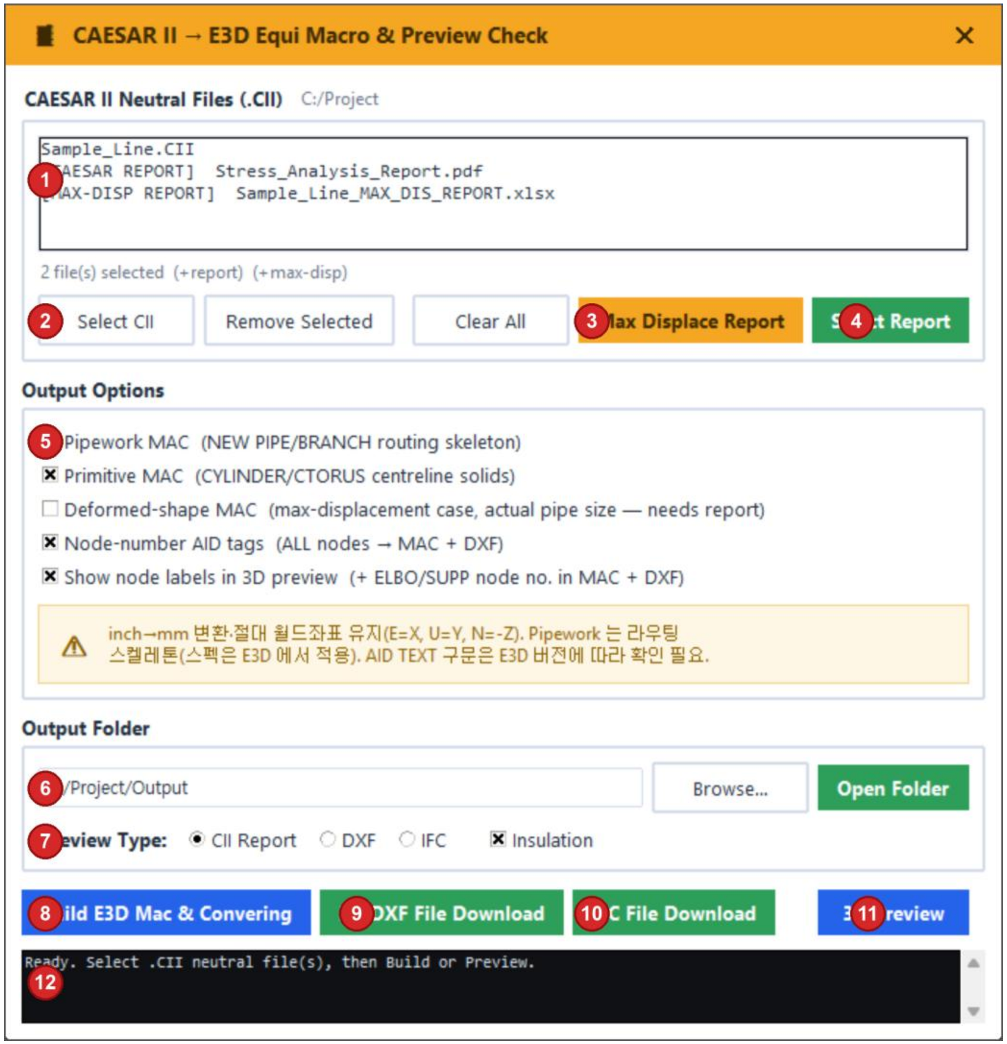
Key Purpose

Why this program

1. **View ALL data from the .CII file alone**
   – No CAESAR II license/program needed — read geometry, supports, sizes directly
2. **Convert everything to 3D design formats**
   – E3D Equipment (MAC) · 3D DXF · SP3D IFC — the whole model, per system
3. **Build the Max-Displacement (deformed) 3D model**
   – Run CLASH CHECK inside Navisworks · AP3D · E3D · SP3D with the deformed shape

Required inputs : CAESAR II Neutral (.CII) + CAESAR II Report + Max Displacement Report

Main Window – Item Guide



1. **CAESAR II Neutral (.CII) files & directory**
2. **Select / Remove / Clear .CII files**
3. **Max Displace Report (Excel/PDF) – deformed source**
4. **Select CAESAR II Report (PDF)**
5. **Output Options (Pipework / Primitive / Deformed / AID / labels)**
6. **Output Folder (Browse / Open Folder)**
7. **Preview Type (CII / DXF / IFC) + Insulation on/off**
8. **Build E3D Mac & Converting**
9. **3D DXF File Download (Shape + Deform)**
10. **IFC File Download (Shape + Deform, for SP3D)**
11. **3D Preview (CII / DXF / IFC)**
12. **Log – progress & results**

Contents

Overview

View without CAESAR II

■ From the .CII file alone — see all geometry / supports / sizes, no CAESAR II needed

One source → many 3D formats

■ E3D Equipment (MAC) · 3D DXF · SP3D IFC — whole model, split per system

■ Type-specific support shapes; reducer / flange / rigid / expansion distinguished

Deformed model → Clash Check

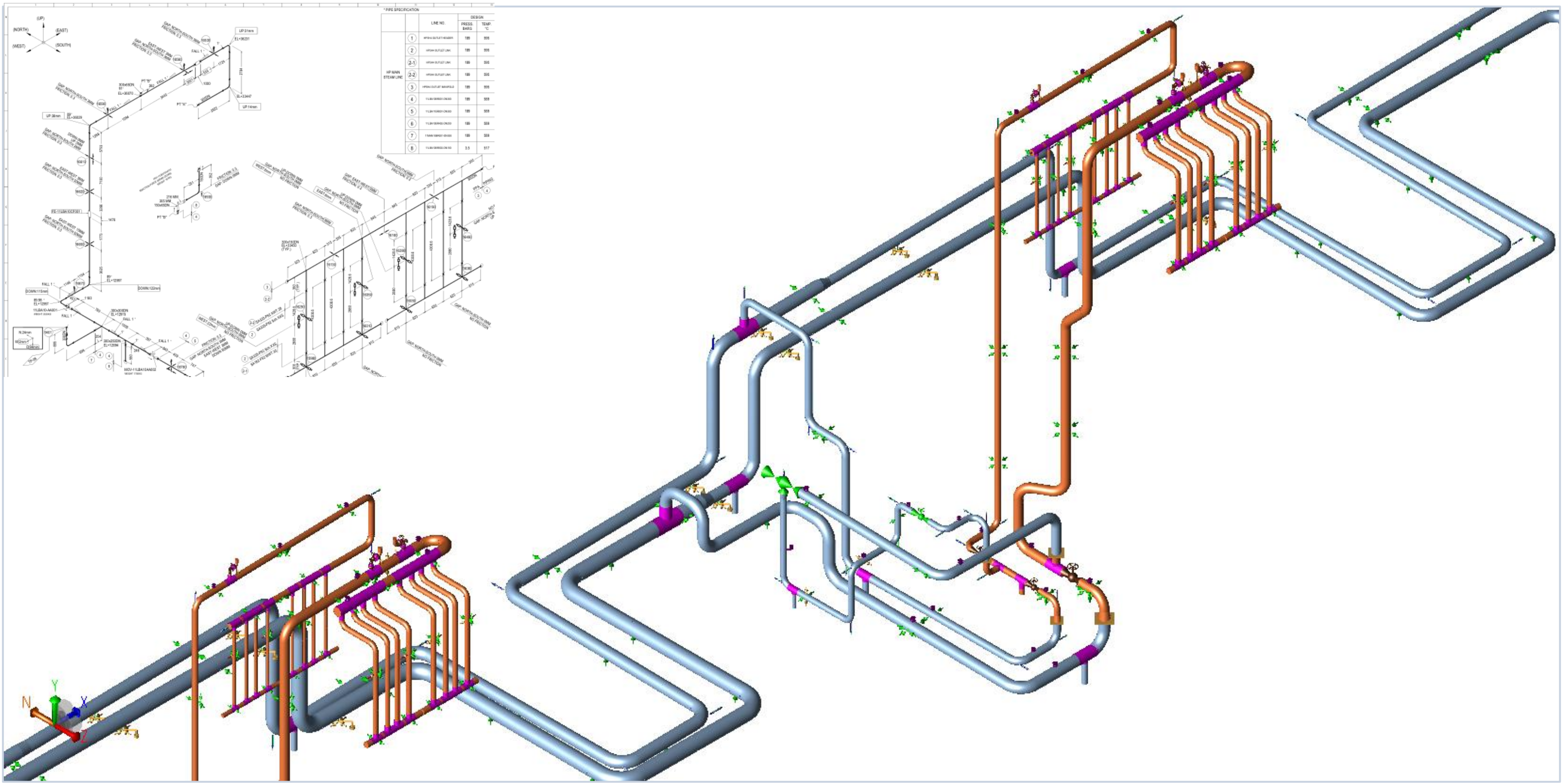
■ Max-displacement (deformed) 3D model for clash check in Navisworks / AP3D / E3D / SP3D

Inputs & Portability

■ Inputs: .CII + CAESAR II Report + Max Displacement Report — offline, pure-Python GUI

Input & Options

1. CAESAR II Neutral (.CII) File



Select one or more .CII files; the directory is shown next to the title.

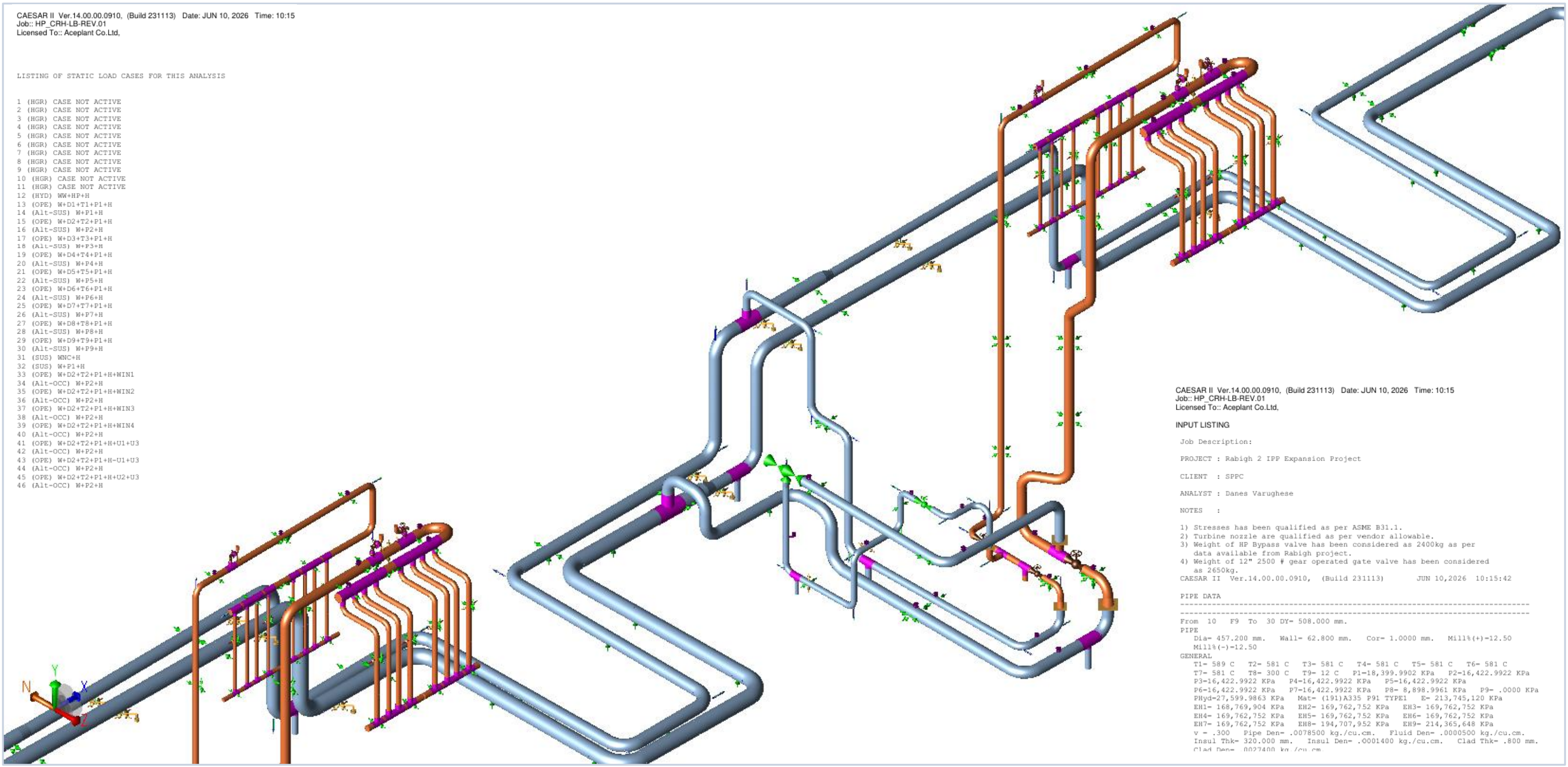
Input & Options

2. Report (Max Displacement / CAESAR Report)

Excel or PDF displacement report drives the deformed-shape output.

Input & Options

3. Input Options Caesar II Report



Choose Pipework / Primitive / Deformed outputs, node labels, Insulation, and Preview Type (CII / DXF / IFC).

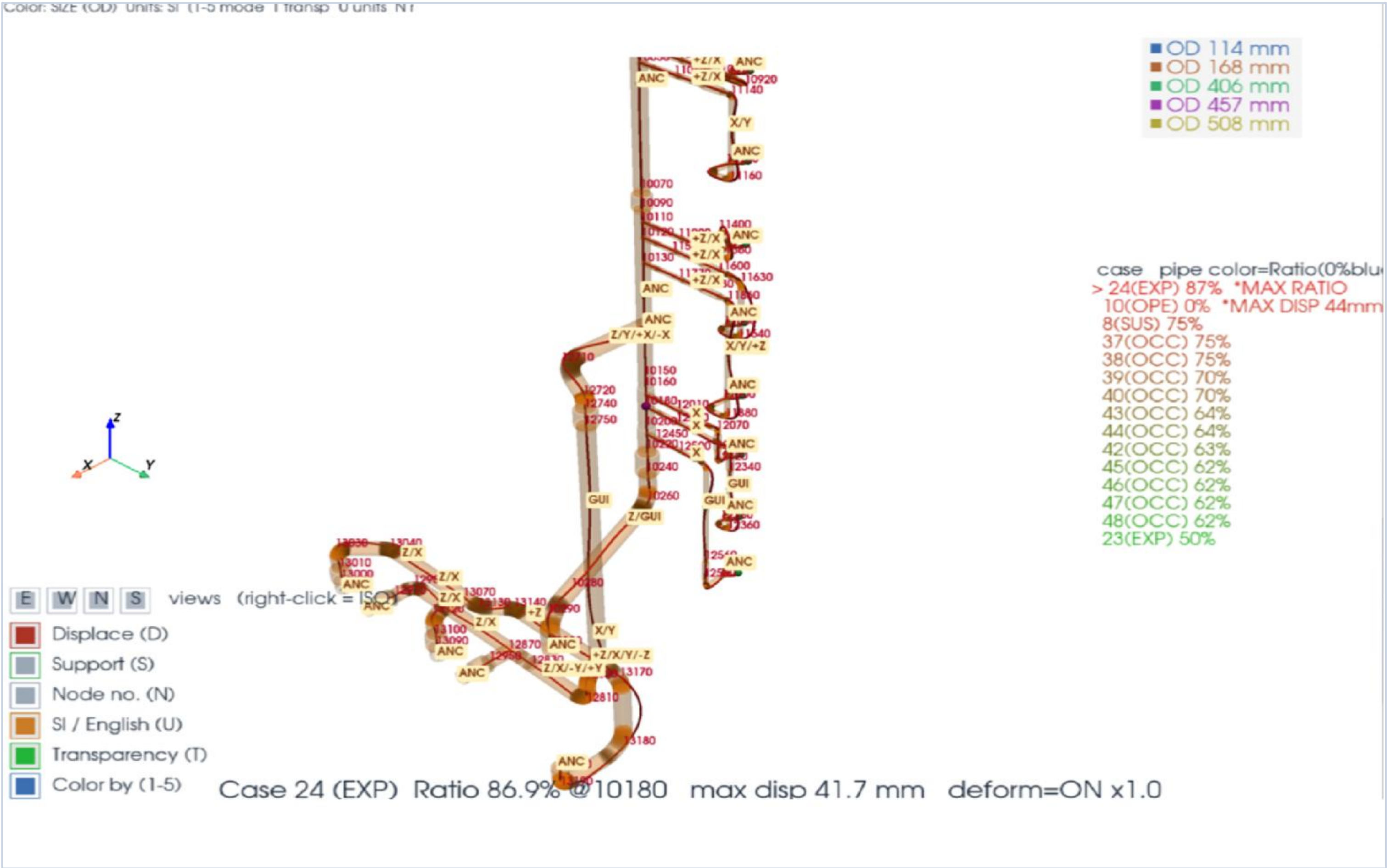
Output – Preview

4-1. 3D Report Data Preview

Preview Type: CII Report / DXF / IFC. Faithful shapes, Shape + Deformed together, node find (F).

Output – Preview

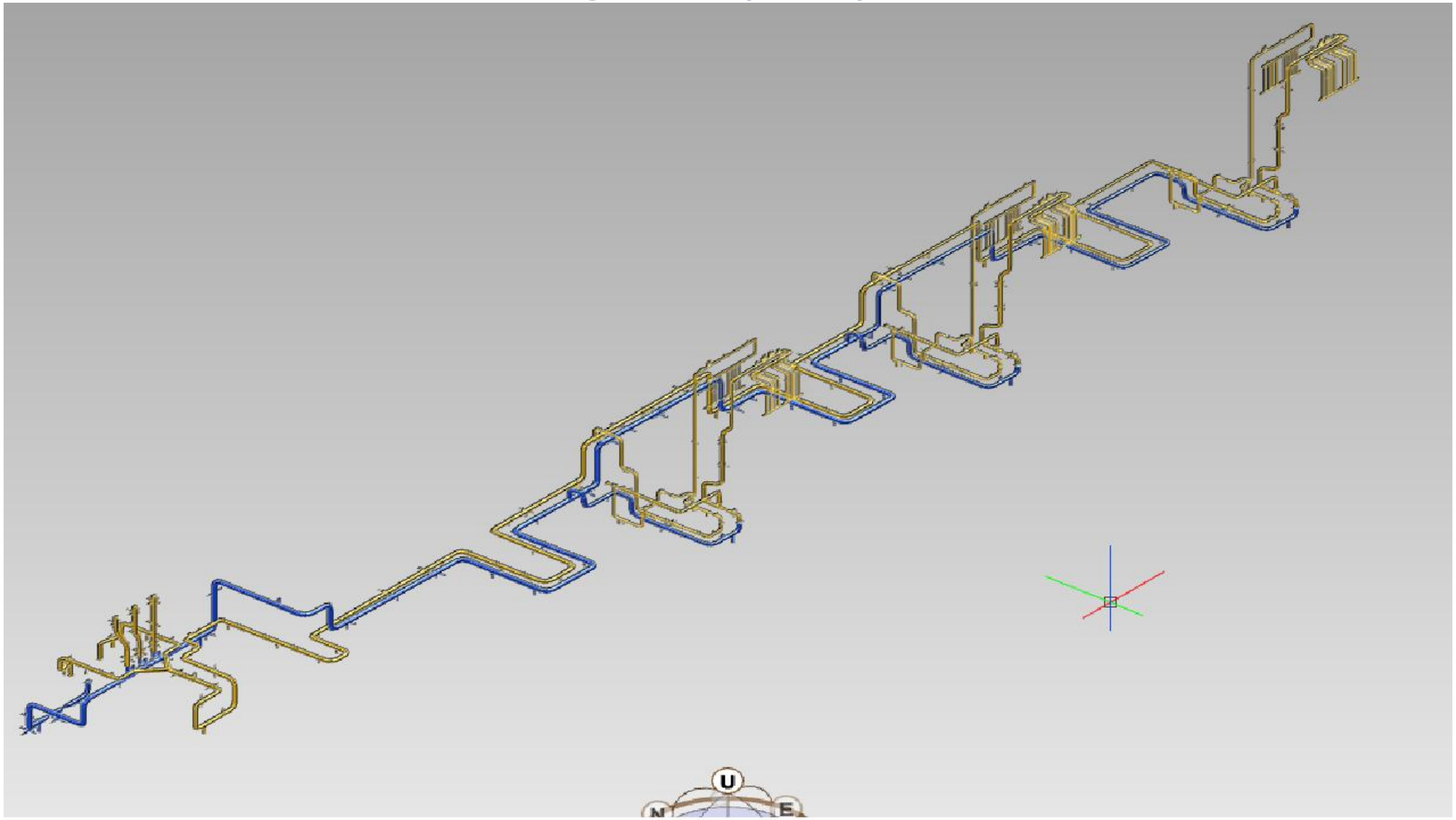
4-2. 3D Report Data Preview



Instant 3D preview: pipes, elbows, supports and node numbers.

Output - E3D Equipment input Macro

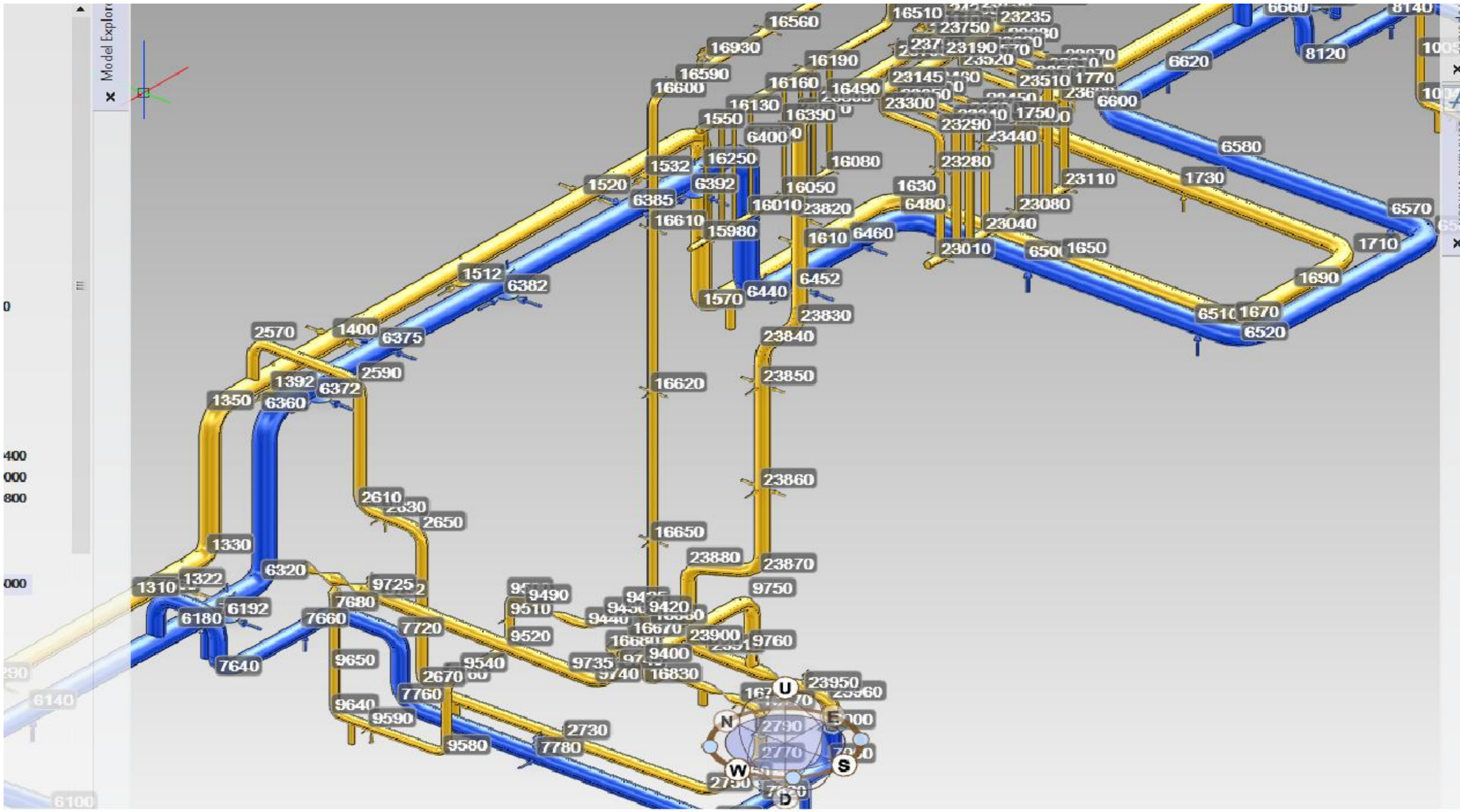
5. E3D Equipment Auto Modeling Input (.MAC)



ZONE > EQUIPMENT (per branch) > SUBE (pipe) + SUBE (each support). Run with $M in E3D.

Output – E3D Clash with Caesar II Movement Data

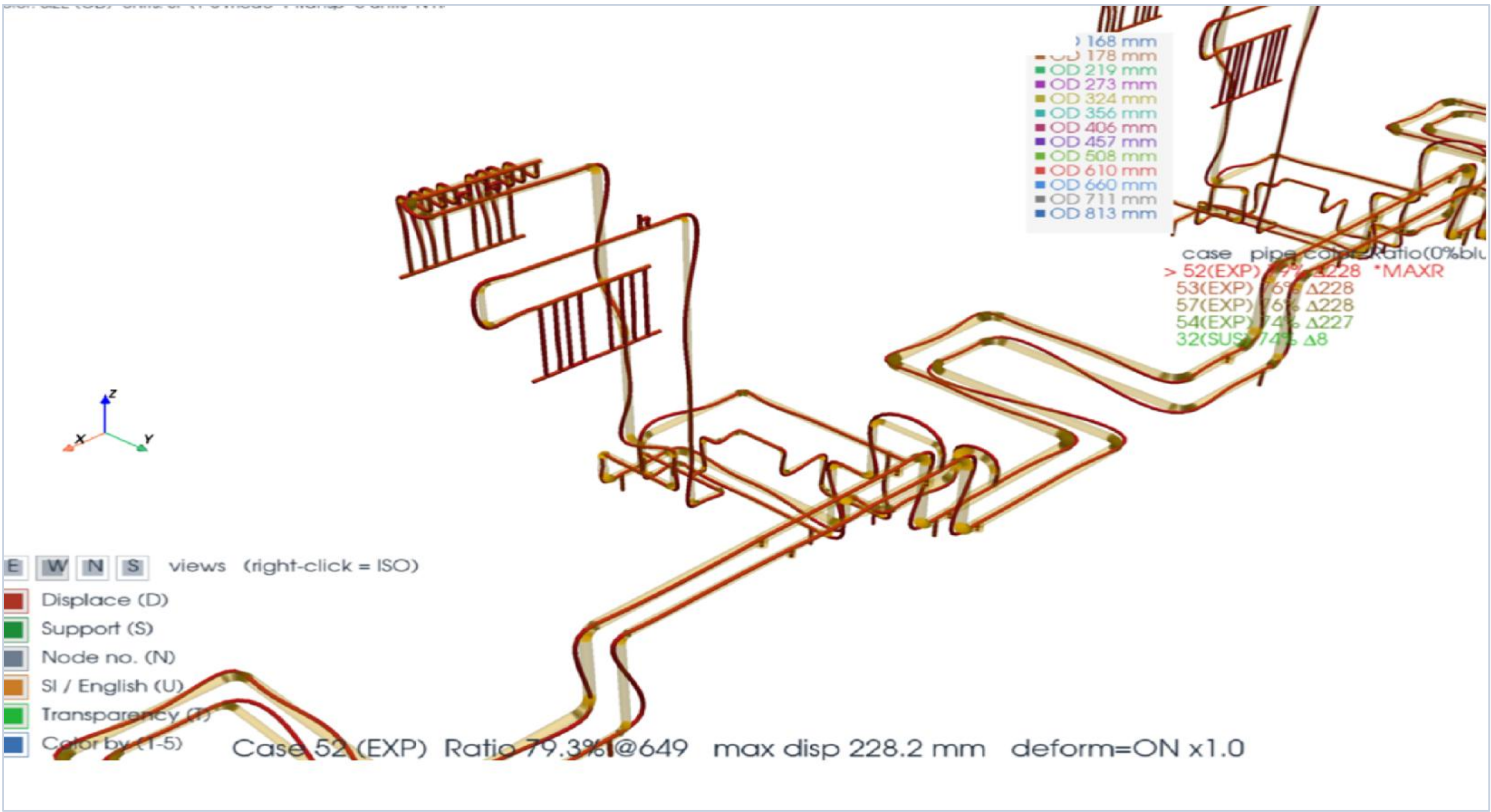
6. Caesar II Shape & Max Disp. Data to E3D Equi Auto Modeling



CYLINDER (pipe) · CTORUS (elbow) · type-specific support solids in E3D.

Output – Movement Data Review

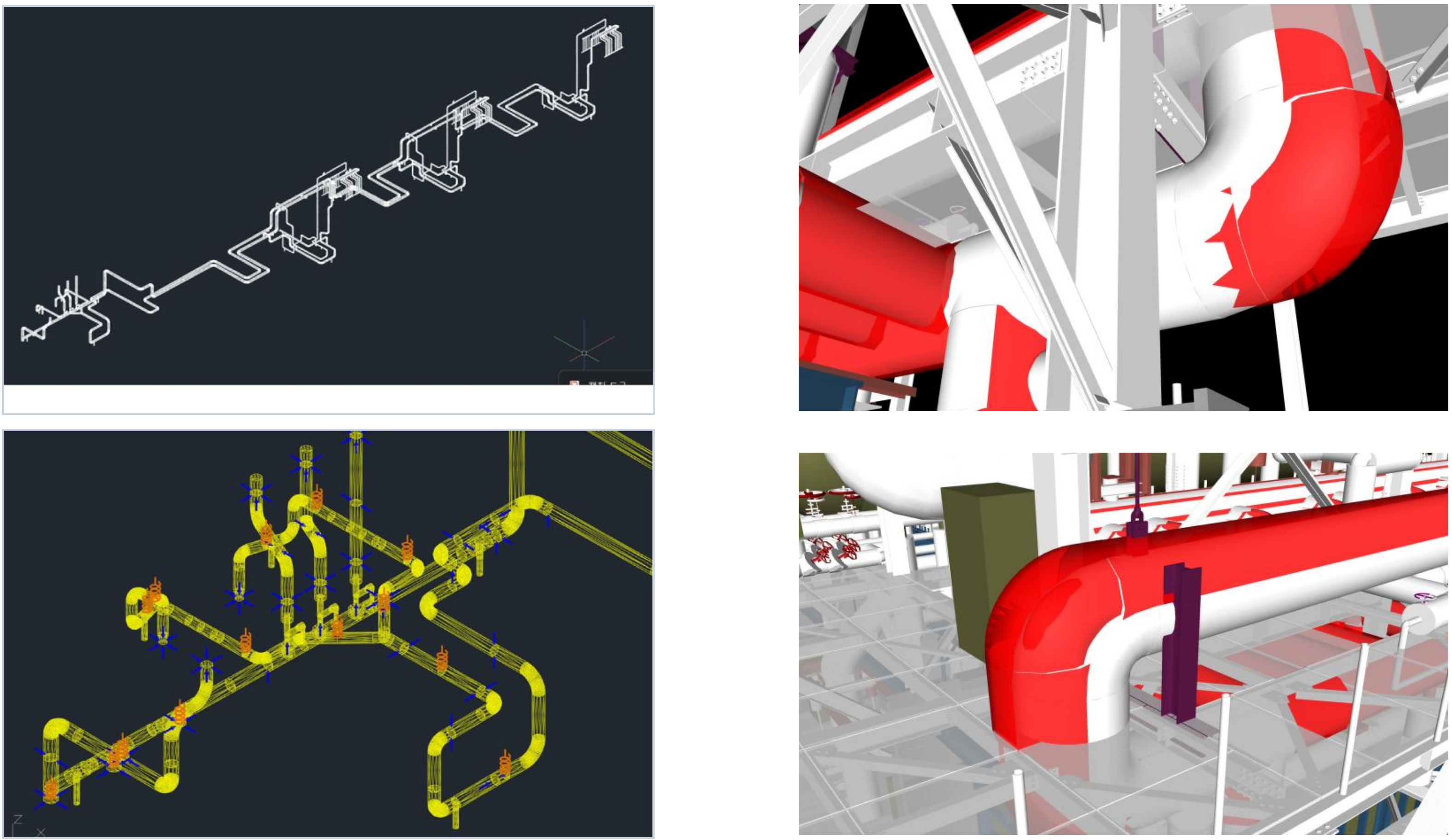
7. Movement Shape Preview → 3D Model Clash Check



Max-displacement deformed model → clash check in Navisworks / AP3D / E3D / SP3D.

Output – 3D DXF for AP3D/NavisWork Review

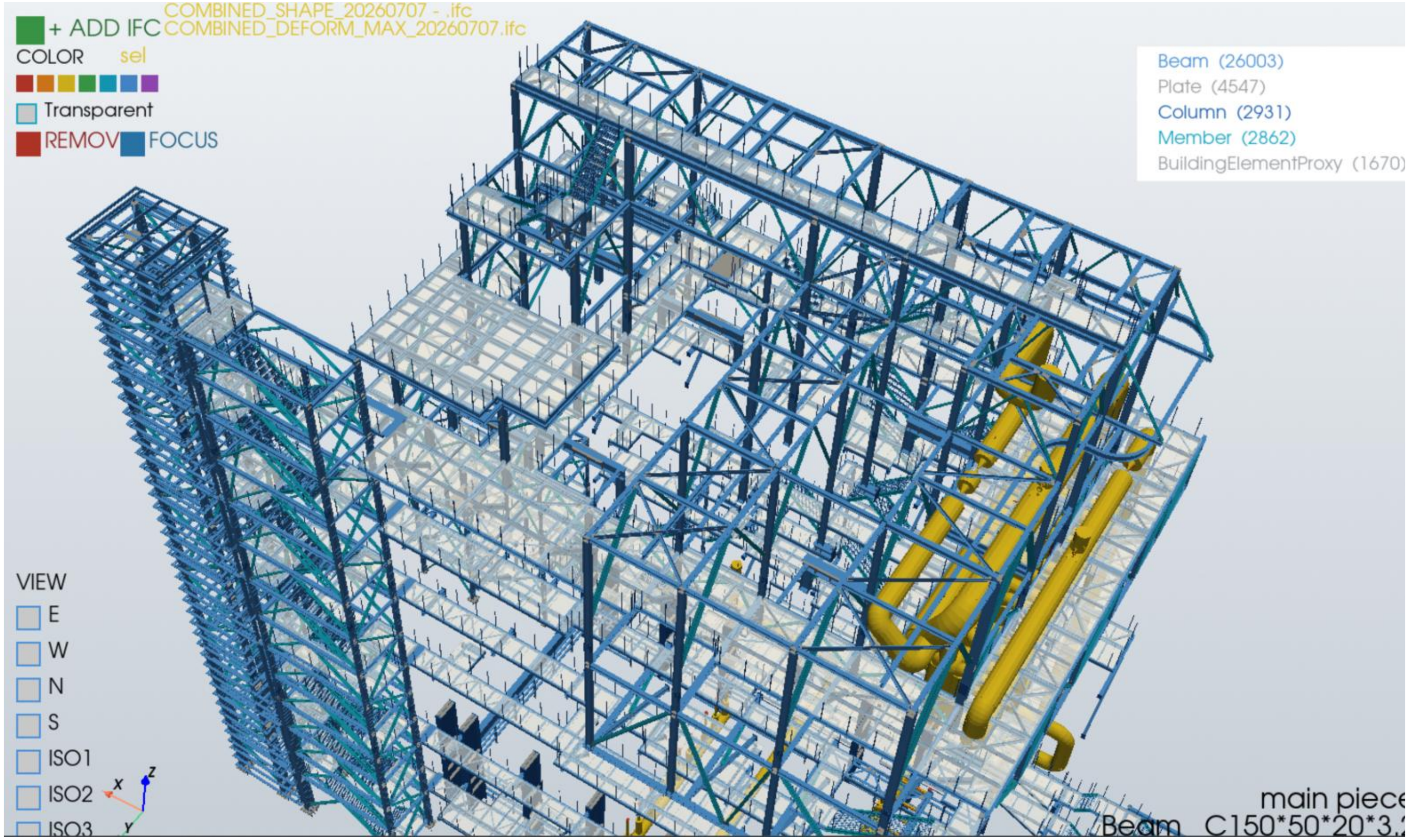
8. 3D DXF (CaesarII Shape & Movement) for Clash Check & Review



3D DXF for CAD / Navisworks / AP3D (no E3D). Layers: PIPE(green) / DEFORM(red) / INSU(gray) / SUPP(blue).

Output – 3D IFC TEKLA

8. 3D IFC (CaesarII Shape & Movement) for Clash Check & Review



IFC solids (IFC2x3) for SP3D via Smart Interop Publisher — verify elbows/reducers before import.

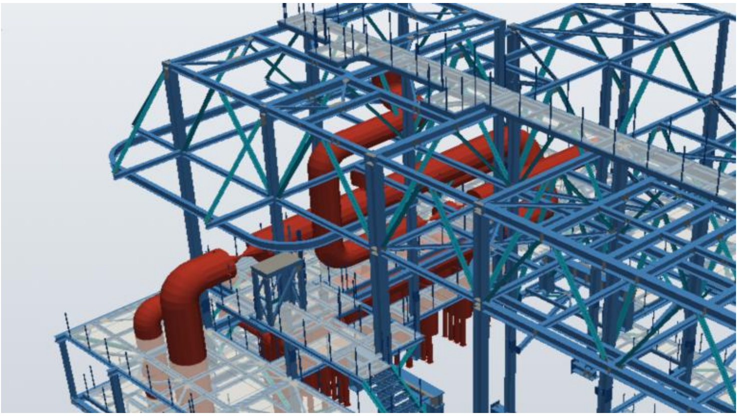
Output – IFC for Smart 3D (SP3D)

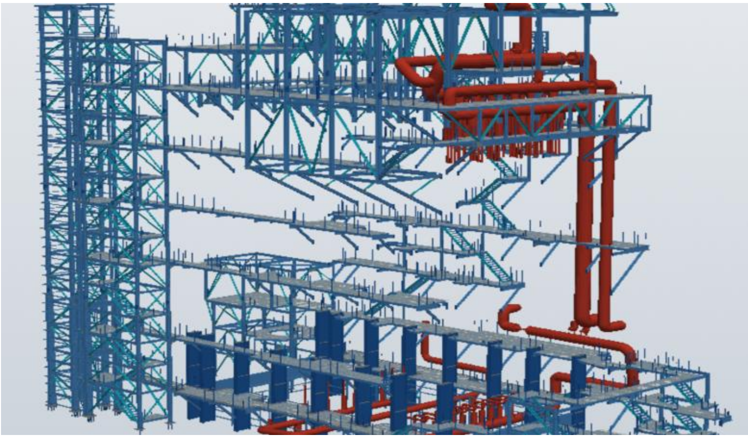
For sites that have SP3D (Smart 3D) but no E3D

- Same verified primitive geometry exported as IFC solids (IFC2x3, SI mm)
- Each branch = IfcElementAssembly; each shape = IfcBuildingElementProxy
- Two files: <name>_SHAPE.ifc + <name>_DEFORM_MAX.ifc

Import route

- Load IFC via Smart Interop Publisher, then reference/review in Smart 3D
- Colored by layer: SHAPE green(PIPE) · DEFORM red · INSU gray · SUPP blue
- Geometry validity verified with ifcopenshell





Support & Component Shapes

Item	Shape
Anchor	Base plate (OD x1.1, 10 mm thick)
Rest / Guide	Direction arrow (gap-aware)
Snubber	Shock absorber (rod + body + base)
Spring hanger	Rod + CTORUS coil (3 rings)
Rod / Limit	Slim rod / axial stop plate
Reducer / Flange	Taper cone / disc pair
Rigid (valve etc.)	Bow-tie cones

Notes & Limitations

- ■ Valves cannot be individually identified — the neutral file has no component-type tag (shown as rigid)
- ■ No line numbers / system names — .CII is node-based (only a few point tags exist)
- ■ E3D MAC AID TEXT / BOX syntax may vary by E3D version — please verify
- ■ IFC is review-grade proxy solids — native piping entities/attributes are future work
- ■ SP3D / SIP import options depend on version (IFC2x3 vs IFC4)

Thank You

CAESAR II → E3D / SP3D · Plant Engineering Total Service System